

CO3

ASSESSMENT TOOL-1

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Tuesday

COURSE CODE : CSA1103

COURSE NAME : OOAD

Experiment 1: Book Bank Management System

Aim

To develop a UML model for implementing the Book Bank Management System that verifies student details using the central computer and issues books after approval.

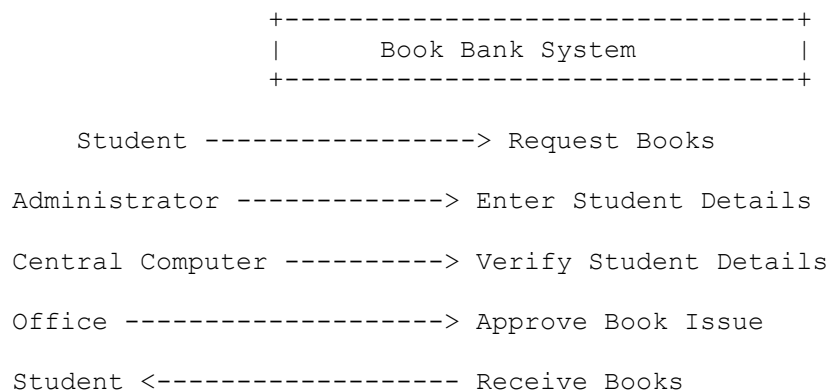
Problem Statement

The student requests books from the Book Bank. The administrator enters the student's details into the central computer. The central computer verifies the details and sends approval to the office. After approval, the office issues the required books to the student.

Actors

- Student
- Administrator
- Central Computer
- Office

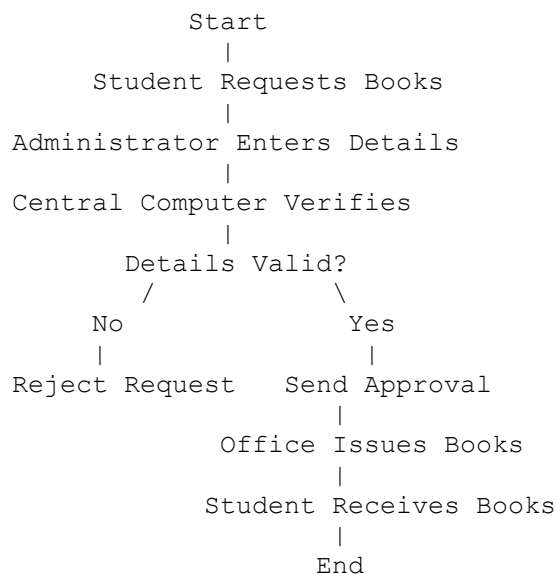
Use Case Diagram



Explanation

- Student requests books.
- Administrator enters student details.
- Central computer verifies the details.
- Office approves the request.
- Books are issued to the student.

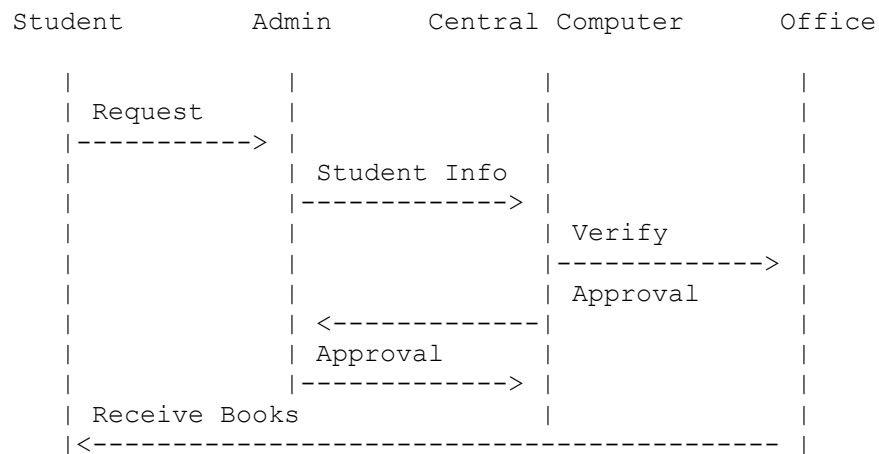
Activity Diagram



Explanation

The process begins with a book request. The administrator submits the student's details for verification. If the student is eligible, approval is granted and the books are issued. Otherwise, the request is rejected.

Sequence Diagram



Explanation

The student requests books. The administrator forwards the details to the central computer. After verification, approval is sent to the office, which issues the books.

Class Diagram

```
+-----+
| Student |
+-----+
| StudentID |
| Name |
| Department |
+-----+
| RequestBook() |
+-----+
```

|

```
+-----+
| Administrator |
+-----+
| AdminID |
| Name |
+-----+
| EnterDetails() |
+-----+
```

|

```
+-----+
| CentralComputer |
+-----+
| StudentDatabase |
+-----+
| VerifyDetails() |
| ApproveRequest() |
+-----+
```

|

```
+-----+
| Office |
+-----+
| OfficeID |
+-----+
| IssueBook() |
+-----+
```

Explanation

- **Student** requests books.
- **Administrator** enters student details.
- **CentralComputer** verifies eligibility.
- **Office** issues books after approval.

Advantages

- Fast verification.
- Reduces paperwork.
- Accurate student validation.
- Easy record management.

Result

The Book Bank Management System successfully verifies student details and issues books using UML-based system design.

Experiment 2: Exam Registration Portal

Aim

To develop a UML model for implementing the **Exam Registration Portal** that verifies candidate details through the central computer and issues the hall ticket after approval.

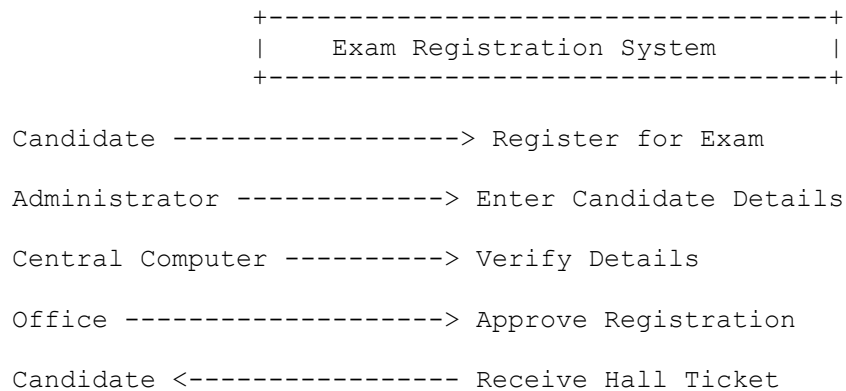
Problem Statement

The candidate registers for the examination. The administrator enters the candidate's details into the central computer. The computer verifies the details and sends approval to the office. After approval, the office issues the hall ticket to the candidate.

Actors

- Candidate
- Administrator
- Central Computer
- Office

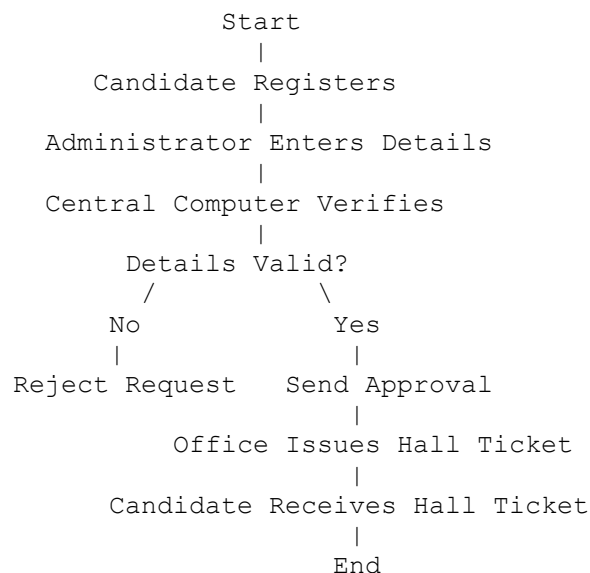
Use Case Diagram



Explanation

- Candidate registers for the exam.
- Administrator enters candidate details.
- Central computer verifies the details.
- Office approves the registration.
- Hall ticket is issued to the candidate.

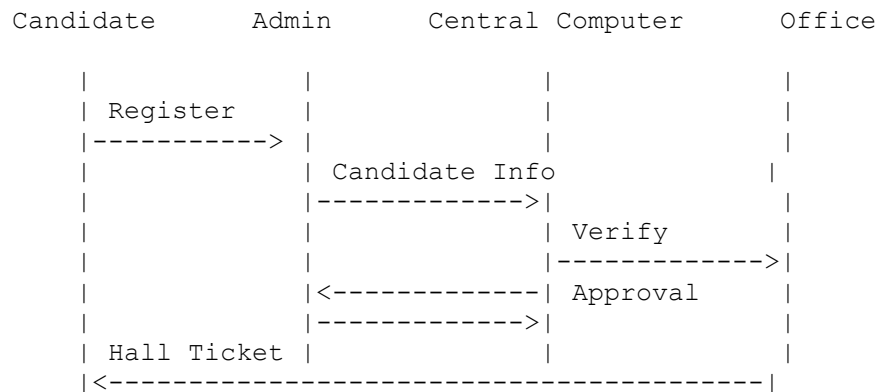
Activity Diagram



Explanation

The process starts with exam registration. After verification, if the details are valid, the hall ticket is generated and issued. Otherwise, the registration is rejected.

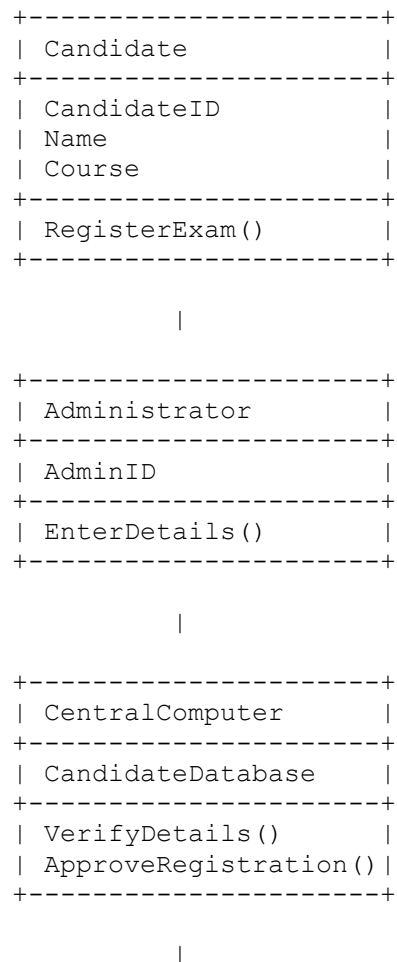
Sequence Diagram



Explanation

The candidate registers. The administrator submits the details to the central computer. After verification, approval is sent to the office, which issues the hall ticket.

Class Diagram



```

+-----+
| Office          |
+-----+
| OfficeID        |
+-----+
| IssueHallTicket() |
+-----+

```

Advantages

- Faster registration process.
- Automatic verification.
- Reduces manual errors.
- Secure hall ticket generation.

Result

The Exam Registration Portal verifies candidate details successfully and issues the hall ticket after approval.

Experiment 3: Stock Maintenance System

Aim

To develop a UML model for implementing the **Stock Maintenance System** for order processing, item delivery, and stock updates.

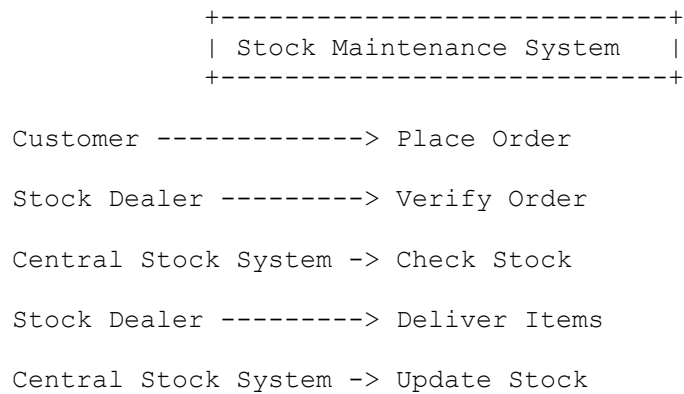
Problem Statement

The customer places an order through the stock dealer. The dealer verifies the order with the central stock system. If the items are available, they are delivered to the customer and the stock quantity is updated automatically.

Actors

- Customer
- Stock Dealer
- Central Stock System

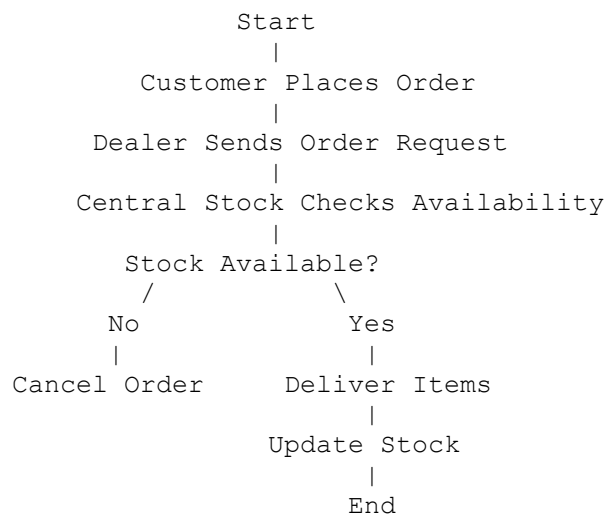
Use Case Diagram



Explanation

The customer places an order. The dealer checks item availability using the stock system. If available, the items are delivered and the stock is updated.

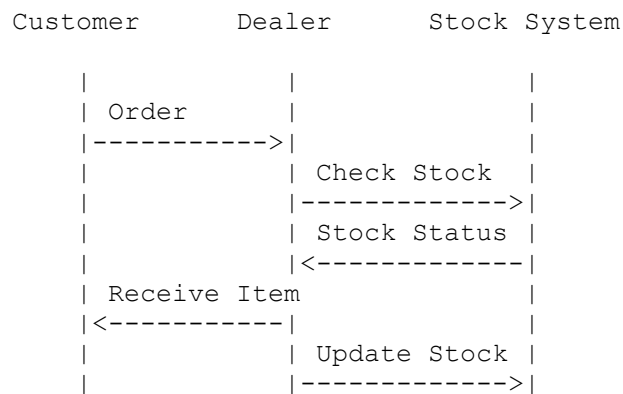
Activity Diagram



Explanation

The system checks stock availability. If the product is available, it is delivered and the inventory is updated. Otherwise, the order is cancelled.

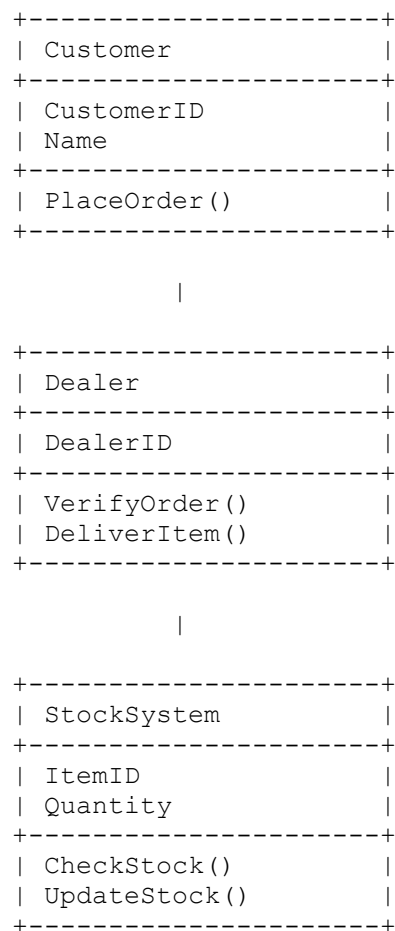
Sequence Diagram



Explanation

The customer places an order. The dealer checks stock availability with the central stock system. After delivery, the stock records are updated.

Class Diagram



Advantages

- Automatic inventory management.
- Faster order processing.
- Accurate stock records.
- Reduces manual work.

Result

The Stock Maintenance System processes orders, delivers items, and updates stock successfully.

Experiment 4: Online Course Reservation System

Aim

To develop a UML model for implementing an **Online Course Reservation System** that allows students to reserve courses based on seat availability.

Problem Statement

The student browses available courses and selects a course. The university checks seat availability through the central management system. If seats are available, the student is enrolled in the selected course.

Actors

- Student
- University
- Central Management System

Use Case Diagram

```
+-----+
| Online Course Reservation System |
+-----+
```

Student -----> Browse Courses

Student -----> Select Course

Management System ---> Check Seat Availability

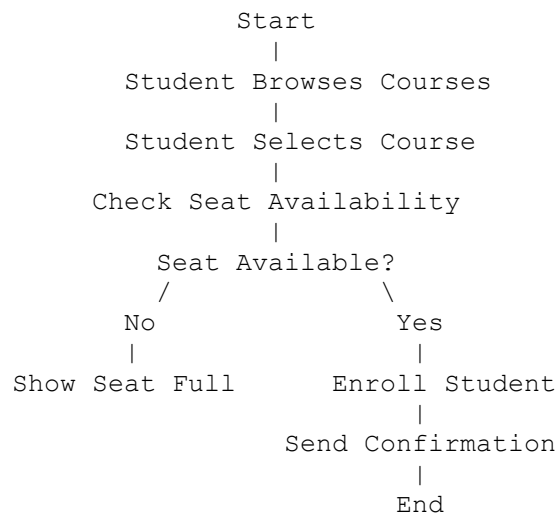
University -----> Approve Enrollment

Student <----- Enrollment Confirmation

Explanation

The student browses and selects a course. The management system checks seat availability. If seats are available, the university enrolls the student.

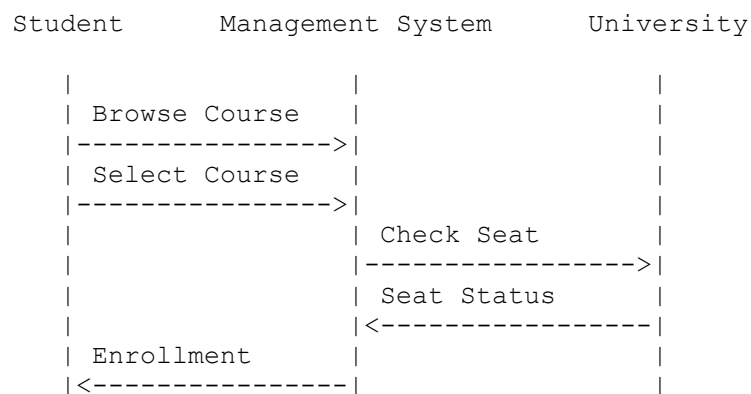
Activity Diagram



Explanation

The system checks whether seats are available for the selected course. If available, the student is enrolled and receives confirmation.

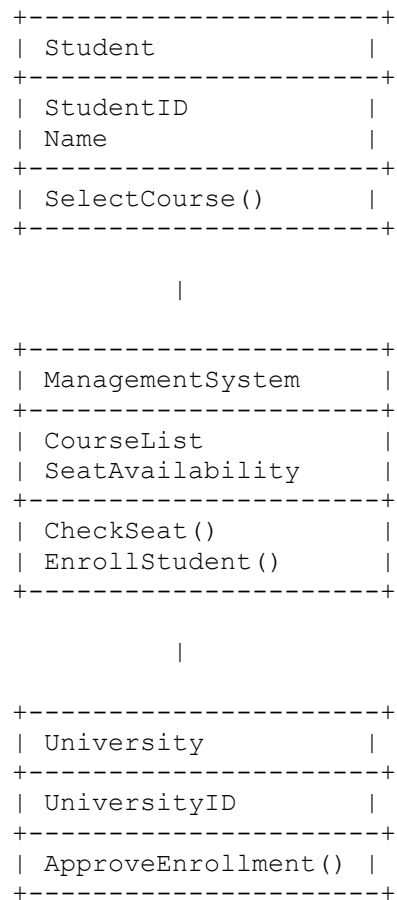
Sequence Diagram



Explanation

The student selects a course. The management system checks seat availability with the university. If seats are available, enrollment is confirmed.

Class Diagram



Advantages

- Easy course registration.
- Automatic seat checking.
- Quick enrollment process.
- Reduces paperwork.

Result

The Online Course Reservation System successfully checks seat availability and enrolls students using UML-based design.